END-TO-END SQL PROJECT — iTunes Music Store Analysis

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1. Project Title

Apple iTunes Music Store Data Analysis Using SQL

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2. Project Objective

The objective of this project is to perform complete end-to-end data analysis using SQL on a relational music store database. The project demonstrates how raw structured data can be transformed into meaningful business insights through querying, aggregation, filtering, and relational joins.

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3. Problem Statement

Businesses collect large amounts of transactional data but cannot use it effectively without analysis. This project aims to analyze customer purchases, revenue trends, product performance, and operational efficiency to support business decision-making.

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4. Dataset Description

The database consists of multiple related tables:

- **Artist** — artist details
- **Album** — albums created by artists
- **Track** — songs available for purchase
- **Genre** — music categories
- **Media_Type** — file formats
- **Customer** — customer information
- **Employee** — staff data
- **Invoice** — purchase transactions

- **Invoice_Line** — items in each purchase
- **Playlist** — curated song lists
- **Playlist_Track** — track mapping

The database is fully relational using primary and foreign keys.

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5. Database Design

Relationships:

- Artist → Album → Track
- Track → Genre
- Track → Media Type
- Customer → Invoice
- Invoice → Invoice Line → Track
- Employee → Customer

This structure ensures normalized data storage and avoids redundancy.

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6. Tools Used

- MySQL
- CSV datasets
- SQL queries
- Relational database concepts

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7. Data Import Process

Steps performed:

1. Created database schema
2. Created tables with constraints
3. Loaded CSV files
4. Handled errors:
    - date format issues
    - foreign key violations
    - null values
    - incorrect data types
5. Verified data integrity

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8. SQL Analysis Performed

Customer Analysis

- Identified top spending customers
- Classified repeat vs one-time buyers
- Calculated customer lifetime value

Sales Analysis

- Monthly revenue trends
- Peak sales months
- Average invoice value

Product Analysis

- Most purchased tracks
- Albums generating highest revenue
- Tracks never purchased

Artist & Genre Analysis

- Top performing artists
- Genre popularity by sales
- Genre demand by region

Employee Performance

- Revenue managed per employee
- Customers handled per representative
- Highest contributing support agents

Geographic Analysis

- Revenue by country
- Customer distribution by city
- High-customer low-revenue regions

Purchase Behavior Analysis

- Purchase frequency distribution
- Average time between purchases
- Multi-genre listening patterns

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9. Key Insights

- A small percentage of customers generate majority of revenue.
- Repeat customers contribute significantly more sales.
- A few artists dominate total revenue.
- Some tracks are never purchased.
- Revenue is concentrated in specific regions.
- Certain genres consistently outperform others.

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10. Business Recommendations

Customer Strategy

- Reward repeat buyers with loyalty programs
- Re-engage inactive customers

Marketing Strategy

- Promote underperforming tracks
- Target high-potential regions

Product Strategy

- Invest in top genres and artists
- Bundle frequently purchased tracks

Operations Strategy

- Balance customer distribution among employees
- Train low-performing representati

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13. Conclusion

This end-to-end SQL project successfully converts structured relational data into meaningful business insights. Through data cleaning, query development, and analytical interpretation, the project demonstrates how SQL can be used as a powerful tool for business intelligence and decision support.

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14. Project Outcome

✓ Database created successfully
✓ Data imported and validated
✓ Queries executed without errors
✓ Insights generated
✓ Business recommendations provided

Project Status: Completed

15. Skills Gained

- SQL querying proficiency
- Database design understanding
- Analytical thinking
- Data interpretation
- Debugging queries
- Problem solving

Final Statement

This project reflects practical industry-level SQL analysis skills and demonstrates the ability to work with real relational datasets to extract actionable insights.